

Chapter 3 - Dimension Formulas

Sections 3.3-3.4: Meromorphic Differentials, Divisors, and Riemann-Roch

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Reference: A First Course in Modular Forms (Diamond and Shurman)

Motivation

Before we look into differentials and Riemann-Roch to see their applications to modular forms, let us take a look at why we are so motivated to use them. Throughout, Γ is a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$. Although there is no discussion of this in the text, many of the statements in this chapter can be relaxed to the assumption that Γ is only a finite index subgroup of $\mathrm{SL}_2(\mathbb{Z})$. For our purposes, we will stick with the stricter condition.

Consider a function f which is weakly modular (for instance, automorphic) of weight k with respect to Γ . Then for $\tau \in \mathcal{H}$ and $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ we know

$$f(\gamma(\tau)) = j(\gamma, \tau)^k f(\tau) = (c\tau + d)^k f(\tau).$$

Previously, in Section 1.1 of the text, we computed

$$d(\gamma(\tau)) = (c\tau + d)^{-2} d\tau = j(\gamma, \tau)^{-2} d\tau.$$

Notice what this says in weight 2: the condition $f(\gamma(\tau)) d(\gamma(\tau)) = f(\tau) d\tau$ is *exactly* the condition $f(\gamma(\tau)) = (c\tau + d)^2 f(\tau)$. More generally, let k be a positive even integer and f weakly modular of weight k . Given $f(\tau)(d\tau)^{k/2}$ we see

$$\begin{aligned} f(\gamma(\tau)) (d\gamma(\tau))^{k/2} &= j(\gamma, \tau)^k f(\tau) \left(j(\gamma, \tau)^{-2} d\tau \right)^{k/2} \\ &= j(\gamma, \tau)^k f(\tau) j(\gamma, \tau)^{-k} (d\tau)^{k/2} \\ &= f(\tau) (d\tau)^{k/2}. \end{aligned}$$

This demonstrates that $f(\tau)(d\tau)^{k/2}$ is Γ -invariant. The goal of this section is to make sense of these differentials, and to show that the automorphic forms of even weight k are naturally isomorphic (as $\mathbb{C}$ -vector spaces) to the meromorphic differentials of degree $k/2$ on $X(\Gamma)$. Thus, the need to understand these differentials and the use of theorems in geometry such as Riemann-Roch is, in essence, to help us understand the subspaces of automorphic forms (which include the spaces of modular forms and cusp forms).

Meromorphic Differentials (Section 3.3)

Local Meromorphic Differentials

We begin with the main definition of this section.

Definition. Let $V \subset \mathbb{C}$ be open and $n \in \mathbb{N}$. The **meromorphic differentials** on V of degree n are

$$\Omega^{\otimes n}(V) = \{f(q)(dq)^n : f \text{ meromorphic on } V\},$$

where q denotes the variable on V .

A seemingly obvious yet important remark is that one must keep track of which coordinate system one works in while considering meromorphic differentials: we will be looking at Riemann surfaces, which have many local charts, and the name of the variable ($q, z, \tau, \dots$) tells us which chart we are in.

Remark. $\Omega^{\otimes n}(V)$ forms a $\mathbb{C}$ -vector space under the usual operations

$$f(q)(dq)^n + g(q)(dq)^n = (f+g)(q)(dq)^n \quad \text{and} \quad c(f(q)(dq)^n) = (cf)(q)(dq)^n \quad \text{for } c \in \mathbb{C}.$$

Moreover, the space of all meromorphic differentials on V ,

$$\Omega(V) = \bigoplus_{n \in \mathbb{N}} \Omega^{\otimes n}(V),$$

forms a ring once we add the multiplication rule $(dq)^n(dq)^m = (dq)^{n+m}$.

The Pullback

Notice the initial definition is only local (it lives on an open subset of $\mathbb{C}$). We would like a way to construct a *globally* defined meromorphic differential on a Riemann surface X . To do this, we need to patch together local meromorphic differentials, meaning we need to understand what happens under the transition maps between local coordinates.

Definition. Let $\varphi : V_1 \rightarrow V_2$ be a holomorphic map between open subsets of $\mathbb{C}$. Then φ induces a **pullback** map $\varphi^* : \Omega^{\otimes n}(V_2) \rightarrow \Omega^{\otimes n}(V_1)$ given by

$$\varphi^*(f(q_2)(dq_2)^n) = f(\varphi(q_1)) \varphi'(q_1)^n (dq_1)^n.$$

The way to remember this formula is that the pullback is nothing more than *substitution*: we substitute $q_2 = \varphi(q_1)$ and $dq_2 = \varphi'(q_1) dq_1$. Note that $f \circ \varphi$ is meromorphic since a meromorphic function composed with a (nonconstant) holomorphic function is

meromorphic, so φ^* really does land in $\Omega^{\otimes n}(V_1)$.

Some important properties of the pullback can be verified straight from the definition. First, **the pullback is linear**: $\varphi^*(\omega_1 + \omega_2) = \varphi^*(\omega_1) + \varphi^*(\omega_2)$ and $\varphi^*(c\omega) = c\varphi^*(\omega)$. (It even respects the ring structure, since $f(\varphi)\varphi'^n \cdot g(\varphi)\varphi'^m = (fg)(\varphi)\varphi'^{n+m}$.) Second, **the pullback is contravariant**, meaning it reverses the order of composition.

Proposition (Contravariance). *Let $\varphi_1 : V_1 \rightarrow V_2$ and $\varphi_2 : V_2 \rightarrow V_3$ be holomorphic maps between open sets $V_1, V_2, V_3 \subset \mathbb{C}$. Then*

$$(\varphi_2 \circ \varphi_1)^* = \varphi_1^* \circ \varphi_2^* : \Omega^{\otimes n}(V_3) \rightarrow \Omega^{\otimes n}(V_1).$$

Proof. Notice that $\varphi_2 \circ \varphi_1$ is a holomorphic map from V_1 to V_3 . We keep track of the substitutions

$$q_2 = \varphi_1(q_1), \quad dq_2 = \varphi_1'(q_1) dq_1, \quad q_3 = \varphi_2(q_2), \quad dq_3 = \varphi_2'(q_2) dq_2.$$

By the definition of the pullback and the chain rule, for $f(q_3)(dq_3)^n \in \Omega^{\otimes n}(V_3)$ we have

$$\begin{aligned} (\varphi_2 \circ \varphi_1)^*(f(q_3)(dq_3)^n) &= f(\varphi_2(\varphi_1(q_1))) \left((\varphi_2 \circ \varphi_1)'(q_1) \right)^n (dq_1)^n \\ &= f(\varphi_2(\varphi_1(q_1))) \left(\varphi_2'(\varphi_1(q_1)) \varphi_1'(q_1) \right)^n (dq_1)^n \\ &= \varphi_1^* \left(f(\varphi_2(q_2)) \varphi_2'(q_2)^n (dq_2)^n \right) \\ &= \varphi_1^* \left(\varphi_2^* (f(q_3)(dq_3)^n) \right). \end{aligned} \quad \square$$

Two ideas were used in the proof: a meromorphic function composed with a holomorphic function is meromorphic, and the chain rule for holomorphic functions. There are a few other properties that come straight from the definition.

Proposition. *If $V_1 \subset V_2$ and $i : V_1 \rightarrow V_2$ is the inclusion map, then $i^*(\omega) = \omega|_{V_1}$ for all $\omega \in \Omega^{\otimes n}(V_2)$.*

Proof. Write $\omega = f(q_2)(dq_2)^n$. Since $i(q_1) = q_1$ for $q_1 \in V_1$, we have $i'(q_1) = 1$, so

$$i^*(\omega) = f(i(q_1)) i'(q_1)^n (dq_1)^n = f(i(q_1)) (dq_1)^n = f(q_1) (dq_1)^n \quad \text{on } V_1,$$

which is exactly $\omega|_{V_1}$. $\square$

In particular (taking $V_1 = V_2$), the identity map pulls back to the identity: $\text{id}^* = \text{id}$.

Proposition. *If $\varphi : V_1 \rightarrow V_2$ is a holomorphic bijection, then $(\varphi^{-1})^* = (\varphi^*)^{-1}$.*

Proof. Recall from complex analysis that a holomorphic bijection has a holomorphic inverse, so $\varphi^{-1} : V_2 \rightarrow V_1$ is holomorphic and $(\varphi^{-1})^* : \Omega^{\otimes n}(V_1) \rightarrow \Omega^{\otimes n}(V_2)$ makes sense. By contravariance and the previous proposition,

$$\varphi^* \circ (\varphi^{-1})^* = (\varphi^{-1} \circ \varphi)^* = (\text{id}_{V_1})^* = \text{id} \quad \text{and} \quad (\varphi^{-1})^* \circ \varphi^* = (\varphi \circ \varphi^{-1})^* = (\text{id}_{V_2})^* = \text{id}.$$

Hence $(\varphi^{-1})^*$ is a two-sided inverse of φ^* . Concretely,

$$(\varphi^*)^{-1}(f(q_1)(dq_1)^n) = f(\varphi^{-1}(q_2))(\varphi^{-1})'(q_2)^n(dq_2)^n = (\varphi^{-1})^*(f(q_1)(dq_1)^n). \quad \square$$

Proposition. *If $\pi : V_1 \rightarrow V_2$ is a holomorphic surjection (with V_1 connected), then π^* is injective.*

Proof. By linearity it suffices to show the kernel is trivial. We have

$$\ker(\pi^*) = \left\{ f(q_2)(dq_2)^n : f(\pi(q_1))\pi'(q_1)^n(dq_1)^n = 0 \right\}.$$

Suppose $f(q_2)(dq_2)^n \in \ker(\pi^*)$ with $f \not\equiv 0$. The zeros and poles of f are isolated, and π is surjective, so $f \circ \pi$ is nonzero on some nonempty open set. There we must have $\pi'(q_1)^n = 0$, so by the identity theorem $\pi' \equiv 0$ on V_1 . Then π is constant, which means π is not surjective onto the open set V_2 , a contradiction. So $f \equiv 0$ and $\ker(\pi^*) = \{0\}$, i.e. π^* is injective. $\square$

Meromorphic Differentials on a Riemann Surface

Equipped with these tools, we can define a global meromorphic differential on a general Riemann surface.

Definition. Let X be a Riemann surface with charts $\varphi_j : U_j \rightarrow V_j$, where $U_j \subset X$, $V_j \subset \mathbb{C}$, and $j \in J$ for some indexing set J . For $j, k \in J$ let

$$V_{j,k} = \varphi_j(U_j \cap U_k) \quad \text{and} \quad V_{k,j} = \varphi_k(U_j \cap U_k),$$

and let $\varphi_{k,j} = \varphi_k \circ \varphi_j^{-1}|_{V_{j,k}} : V_{j,k} \rightarrow V_{k,j}$ be the transition map. A **meromorphic differential** on X of degree n is a collection of local meromorphic differentials of degree n ,

$$(\omega_j)_{j \in J} \in \prod_{j \in J} \Omega^{\otimes n}(V_j),$$

that is **compatible**, meaning that for all $j, k \in J$,

$$\varphi_{k,j}^*(\omega_k|_{V_{k,j}}) = \omega_j|_{V_{j,k}}.$$

At first glance, this definition can seem quite daunting. All it is saying is that the local pieces agree once we change coordinates: if we take the differential written in the k th coordinate and rewrite it (pull it back) in the j th coordinate, we get the differential written in the j th coordinate.

Remark. Much like local meromorphic differentials, the set $\Omega^{\otimes n}(X)$ of meromorphic differentials of degree n on X is a $\mathbb{C}$ -vector space, and

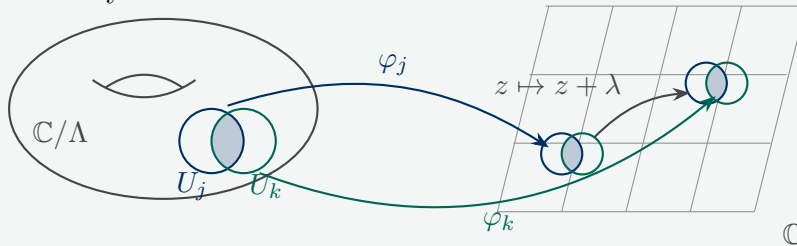
$$\Omega(X) = \bigoplus_{n \in \mathbb{N}} \Omega^{\otimes n}(X)$$

forms a ring. (Compatibility is preserved by sums, scalar multiples and products because each pullback $\varphi_{k,j}^*$ is linear and multiplicative.)

Let's work through an example to see how to apply such a definition.

Example. Take a complex torus $X = \mathbb{C}/\Lambda$, where Λ is a lattice in $\mathbb{C}$ (recall a complex torus can be identified with a complex elliptic curve). We have the natural projection $\pi : \mathbb{C} \rightarrow \mathbb{C}/\Lambda$, and we take the charts $\varphi_j : U_j \rightarrow V_j$ on the torus to be the local homeomorphic inverses of the natural projection $\pi : V_j \rightarrow U_j$.

Fact: every transition map takes the form $z \mapsto z + \lambda$ for some $\lambda \in \Lambda$ (on each connected piece of the overlap). Geometrically, two overlapping charts land in two copies of the overlap that differ by a lattice translation:



Using the definition above, $V_{j,k}$ and $V_{k,j}$ are the same set modulo Λ . Consider the collection of local meromorphic differentials $\omega = ((dz_j)^n)_{j \in J}$, where z_j is the variable on V_j . Then

$$\varphi_{k,j}^*((dz_k)^n) = \varphi'_{k,j}(z_j)^n (dz_j)^n = (dz_j)^n,$$

since $\varphi'_{k,j}(z_j) = (z_j + \lambda)' = 1$. So the collection is compatible, and we have a globally defined meromorphic differential $(dz)^n$ on the torus. Notice it has no zeros and no poles anywhere; we will come back to this example in Section 3.4.

Meromorphic Differentials on $X(\Gamma)$

Now consider the modular curve $X(\Gamma) = \Gamma \backslash \mathcal{H}^*$ and the natural projection

$$\pi : \mathcal{H} \longrightarrow X(\Gamma), \quad \tau \longmapsto \Gamma\tau.$$

We want to make sense of meromorphic differentials on $X(\Gamma)$. A question one could ask is: what is the pullback map? We would like it to look like

$$\pi^* : \Omega^{\otimes n}(X(\Gamma)) \longrightarrow \Omega^{\otimes n}(\mathcal{H}), \quad \omega \longmapsto f(\tau)(d\tau)^n,$$

where $f(\tau)(d\tau)^n$ should be a Γ -invariant object.

Recall from Chapter 2 that the coordinate neighborhoods on $X(\Gamma)$ are $\{\pi(U_j) : j \in J\}$, where each $U_j \subset \mathcal{H}^*$ is a neighborhood of either a point $\tau_j \in \mathcal{H}$ or a cusp $s_j \in \mathbb{Q} \cup \{\infty\}$. The local coordinate map $\varphi_j : \pi(U_j) \rightarrow V_j \subset \mathbb{C}$ has the property that

$$\psi_j = \varphi_j \circ \pi : U_j \longrightarrow V_j$$

is a concrete map (identify by Γ , then map out to $\mathbb{C}$). Formally this gives $\psi_j^* = \pi^* \circ \varphi_j^*$, which tells us how to define π^* .

Let $\omega = (\omega_j)_{j \in J} \in \Omega^{\otimes n}(X(\Gamma))$ and set

$$U'_j = U_j \cap \mathcal{H} \subset \mathcal{H}, \quad V'_j = \psi_j(U'_j) \subset \mathbb{C}, \quad \omega'_j = \omega_j|_{V'_j}.$$

(We remove the cusp, if there is one, so that ψ_j is a holomorphic map between open subsets of $\mathbb{C}$.) Then $\pi^*(\omega) \in \Omega^{\otimes n}(\mathcal{H})$ is *locally defined* by

$$\pi^*(\omega)|_{U'_j} = \psi_j^*(\omega'_j) \quad \text{for all } j \in J.$$

Note that this only tells us what the pullback is locally. We want to show this construction gives a single global meromorphic differential $\pi^*(\omega) = f(\tau)(d\tau)^n$ on $\mathcal{H}$. Consider the commutative diagram

$$\begin{array}{ccccc} & & U_j \cap U_k & & \\ & \swarrow \psi_j & \downarrow \pi & \searrow \psi_k & \\ V_{j,k} & \xleftarrow{\varphi_j} & \pi(U_j \cap U_k) & \xrightarrow{\varphi_k} & V_{k,j} \\ & & \searrow \varphi_{k,j} & & \end{array}$$

The transition map is $\varphi_{k,j} = \varphi_k \circ \varphi_j^{-1}|_{V_{j,k}}$, and the diagram commutes, so

$$\varphi_{k,j} \circ \psi_j = \psi_k \quad \text{on } U_j \cap U_k \quad \implies \quad \psi_k^* = \psi_j^* \circ \varphi_{k,j}^*$$

by contravariance. If we let

$$V'_{j,k} = \psi_j(U'_j \cap U'_k) \quad \text{and} \quad V'_{k,j} = \psi_k(U'_j \cap U'_k),$$

then

$$\underbrace{\psi_k^*(\omega_k|_{V'_{k,j}})}_{\pi^*(\omega)|_{U'_j \cap U'_k} \text{ via chart } k} = \psi_j^*\left(\varphi_{k,j}^*(\omega_k|_{V'_{k,j}})\right) = \underbrace{\psi_j^*(\omega_j|_{V'_{j,k}})}_{\pi^*(\omega)|_{U'_j \cap U'_k} \text{ via chart } j},$$

where the last equality is compatibility, since $(\omega_j)_{j \in J} \in \Omega^{\otimes n}(X(\Gamma))$. So the local pullbacks have a common restriction $f(\tau)(d\tau)^n|_{U'_j \cap U'_k}$ on every overlap, and therefore

$$\pi^*(\omega) = f(\tau)(d\tau)^n \quad \text{is well-defined.}$$

Since each ω_j is meromorphic and each ψ_j is holomorphic, f is meromorphic on $\mathcal{H}$.

Remark. Since the pullback comes from an object on $X(\Gamma)$, it must be Γ -invariant. Indeed, $\pi \circ \gamma = \pi$ for every $\gamma \in \Gamma$, so $\gamma^* \circ \pi^* = \pi^*$. Using $\gamma'(\tau) = j(\gamma, \tau)^{-2}$ and $(f[\gamma]_k)(\tau) = j(\gamma, \tau)^{-k} f(\gamma(\tau))$, this reads

$$\begin{aligned} f(\tau)(d\tau)^n &= \gamma^*(f(\tau)(d\tau)^n) = f(\gamma(\tau)) \gamma'(\tau)^n (d\tau)^n \\ &= f(\gamma(\tau)) j(\gamma, \tau)^{-2n} (d\tau)^n \\ &= (f[\gamma]_{2n})(\tau) (d\tau)^n. \end{aligned}$$

Therefore $f[\gamma]_{2n} = f$ for all $\gamma \in \Gamma$, i.e. f is **weakly modular of weight $2n$** . And f is meromorphic at ∞ in the sense that $f[\alpha]_{2n}$ is meromorphic at ∞ for all $\alpha \in \text{SL}_2(\mathbb{Z})$. This second statement is a non-obvious fact, so we prove it.

Proof of the meromorphy at the cusps. Let $\alpha \in \text{SL}_2(\mathbb{Z})$ and let $s = \alpha(\infty)$. Recall from Chapter 2 (pages 49 and 60 of the text) that the local map at the cusp s is $\psi = \rho \circ \delta : U \rightarrow V$, where

$$\delta = \alpha^{-1}, \quad \rho(z) = e^{2\pi iz/h}, \quad h = \text{the width of } s.$$

Since ω is meromorphic at the cusps, $\omega|_V = g(q)(dq)^n$ for some g meromorphic at 0. We compute f on $U - \{s\}$. Using that $\delta \in \text{SL}_2(\mathbb{Z})$ gives $\delta'(\tau) = j(\delta, \tau)^{-2}$,

$$\begin{aligned} \psi^*(\omega|_{V-\{0\}}) &= \delta^*\left(\rho^*(g(q)(dq)^n)\right) \\ &= \delta^*\left(g(\rho(z)) \rho'(z)^n (dz)^n\right) \\ &= g(\rho(\delta(\tau))) \rho'(\delta(\tau))^n \delta'(\tau)^n (d\tau)^n \\ &= g\left(e^{2\pi i \delta(\tau)/h}\right) \left(\frac{2\pi i}{h} e^{2\pi i \delta(\tau)/h}\right)^n j(\delta, \tau)^{-2n} (d\tau)^n. \end{aligned}$$

Let $q_h = e^{2\pi iz/h}$ and define

$$\tilde{f}(z) = g(q_h) q_h^n \left(\frac{2\pi i}{h} \right)^n.$$

Then the computation above says exactly

$$\psi^*(\omega|_{V-\{0\}}) = \tilde{f}(\delta(\tau)) j(\delta, \tau)^{-2n} (d\tau)^n = (\tilde{f}[\delta]_{2n})(\tau) (d\tau)^n,$$

so

$$f = \tilde{f}[\delta]_{2n} \quad \text{gives} \quad \psi^*(\omega|_{V-\{0\}}) = f(\tau)(d\tau)^n.$$

Since $[\cdot]_{2n}$ is a right action and $\delta\alpha = I$,

$$(f[\alpha]_{2n})(z) = (\tilde{f}[\delta]_{2n}[\alpha]_{2n})(z) = \tilde{f}(z),$$

and $\tilde{f}$ is a meromorphic function of q_h at 0 (it is g times a power of q_h). Hence $f[\alpha]_{2n}$ is meromorphic at ∞ . $\square$

Remark. Every meromorphic differential ω of degree n on $X(\Gamma)$ pulls back to $\pi^*(\omega) = f(\tau)(d\tau)^n$, a meromorphic differential on $\mathcal{H}$, where $f \in \mathcal{A}_{2n}(\Gamma)$ is an automorphic form of weight $2n$ with respect to Γ .

Reversing the Calculation

A natural question to ask after this calculation is whether we can reverse it. That is, can we create an invertible map from the automorphic forms of even weight to the meromorphic differentials on $X(\Gamma)$ of half the weight? The key tool is the following.

Proposition (Exercise 3.3.2). *A collection $(\omega_j)_{j \in J}$ of local meromorphic differentials is compatible (making (ω_j) a meromorphic differential on $X(\Gamma)$) **if and only if** its local elements pull back to restrictions of a single $f(\tau)(d\tau)^n \in \Omega^{\otimes n}(\mathcal{H})$ with $f \in \mathcal{A}_{2n}(\Gamma)$.*

Goal: Given $f \in \mathcal{A}_{2n}(\Gamma)$, construct a meromorphic differential $\omega(f) \in \Omega^{\otimes n}(X(\Gamma))$ such that

$$\pi^*(\omega(f)) = f(\tau)(d\tau)^n.$$

By the proposition, it is enough to construct local differentials ω_j on each V_j that pull back to restrictions of $f(\tau)(d\tau)^n$. Recall the local maps

$$\psi_j = \rho_j \circ \delta_j : \quad \tau \longmapsto z \longmapsto q,$$

which move from τ -space to z -space to q -space. Here $\delta_j \in \text{GL}_2(\mathbb{C})$, so the first step is invertible, while ρ_j (a power map or an exponential) is not invertible in general.

Strategy for Reversing the Calculation

Step 1 (τ -space to z -space). Push $f(\tau)(d\tau)^n$ forward to z -space using the invertible matrix $\alpha = \delta_j^{-1}$, producing a differential λ_j with $\delta_j^*(\lambda_j) = f(\tau)(d\tau)^n|_{U'_j}$.

Step 2 (z -space to q -space). Use the invariance of λ_j to find ω_j on V_j with $\rho_j^*(\omega_j) = \lambda_j$. This splits into two cases: U_j a neighborhood of a point of $\mathcal{H}$, or of a cusp.

Step 3 (*Assemble*). Then $\psi_j^*(\omega_j) = \delta_j^* \rho_j^*(\omega_j) = f(\tau)(d\tau)^n|_{U'_j}$ for every j , and the proposition shows $(\omega_j)_{j \in J}$ is compatible.

Since $\delta_j \in \mathrm{GL}_2(\mathbb{C})$ need not have determinant 1, we first extend the weight- k operator.

Definition. For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{C})$ let $j(\gamma, \tau) = c\tau + d$ and define the **weight- k operator**

$$(f[\gamma]_k)(\tau) = (\det \gamma)^{k/2} j(\gamma, \tau)^{-k} f(\gamma(\tau)).$$

Remark. The exponent on $\det \gamma$ is for normalization. It is chosen so that scalar matrices act trivially in even weight (for $\gamma = cI$, $(\det \gamma)^{k/2} j(\gamma, \tau)^{-k} = c^k c^{-k} = 1$), so that $[\gamma]_k[\gamma']_k = [\gamma\gamma']_k$ still holds, and so that it absorbs the derivative $\gamma'(\tau) = \det(\gamma) j(\gamma, \tau)^{-2}$ when we pull back, as we see next.

Step 1. Let $U'_j = U_j \cap \mathcal{H}$ and $\alpha = \delta_j^{-1}$. Define λ_j to be the pull forward of $f(\tau)(d\tau)^n$ to z -space, which is the pullback by α :

$$\begin{aligned} \lambda_j &= \alpha^* \left(f(\tau)(d\tau)^n|_{U'_j} \right) = f(\alpha(z)) \alpha'(z)^n (dz)^n \\ &= (\det \alpha)^n j(\alpha, z)^{-2n} f(\alpha(z)) (dz)^n \\ &= (f[\alpha]_{2n})(z) (dz)^n. \end{aligned}$$

Since $\alpha\delta_j = I$, contravariance gives $\delta_j^*(\lambda_j) = (\alpha\delta_j)^*(f(\tau)(d\tau)^n) = f(\tau)(d\tau)^n|_{U'_j}$.

Also, λ_j is $\delta_j\Gamma\delta_j^{-1}$ -**invariant**. Let $\gamma \in \Gamma$. By contravariance, $(\delta_j\gamma\delta_j^{-1})^* = (\delta_j^{-1})^* \gamma^* \delta_j^*$, and we compute each piece:

$$\begin{aligned} \delta_j^*(\lambda_j) &= \delta_j^* \left(f(\alpha(z)) (d\alpha(z))^n \right) = f(\alpha(\delta_j(\tau))) \left(d\alpha(\delta_j(\tau)) \right)^n = f(\tau)(d\tau)^n, \\ \gamma^* \left(f(\tau)(d\tau)^n \right) &= f(\tau)(d\tau)^n \quad (\text{since } f[\gamma]_{2n} = f), \\ (\delta_j^{-1})^* \left(f(\tau)(d\tau)^n \right) &= f(\alpha(z)) (d\alpha(z))^n = \lambda_j. \end{aligned}$$

Thus $(\delta_j\gamma\delta_j^{-1})^*(\lambda_j) = \lambda_j$.

Step 2. Now we move from z -space to q -space. The problem is that ρ_j is not invertible in general, so we cannot simply push forward. We treat the two cases separately.

Case 1: $U_j \subset \mathcal{H}$ (not working with cusps)

Here U_j is a neighborhood of a point $\tau_j \in \mathcal{H}$ with $\delta_j(\tau_j) = 0$, and the group

$$\{\pm I\} \delta_j \Gamma_{\tau_j} \delta_j^{-1} / \{\pm I\}$$

(where Γ_{τ_j} is the isotropy subgroup of τ_j) is cyclic of order $h = h_j$, the period of τ_j . It is generated by the rotation

$$r_h : z \mapsto \mu_h z, \quad \mu_h = e^{2\pi i/h},$$

and the local coordinate is $q = \rho_j(z) = z^h$. By the $\delta_j \Gamma_{\tau_j} \delta_j^{-1}$ -invariance of λ_j ,

$$r_h^*(\lambda_j) = (f[\alpha]_{2n})(\mu_h z) \mu_h^n (dz)^n = \lambda_j = (f[\alpha]_{2n})(z) (dz)^n,$$

which holds if and only if

$$\mu_h^n (f[\alpha]_{2n})(\mu_h z) = (f[\alpha]_{2n})(z), \quad \text{or equivalently} \quad (\mu_h z)^n (f[\alpha]_{2n})(\mu_h z) = z^n (f[\alpha]_{2n})(z).$$

So the meromorphic function $z^n (f[\alpha]_{2n})(z)$ is invariant under $z \mapsto \mu_h z$. Writing its Laurent expansion as $\sum_m a_m z^m$, the invariance says $a_m \mu_h^m = a_m$ for all m , so $a_m = 0$ unless $h \mid m$. Hence

$$z^n (f[\alpha]_{2n})(z) = g_j(z^h) \quad \text{for some meromorphic } g_j.$$

Define

$$\omega_j = \frac{g_j(q)}{(hq)^n} (dq)^n \quad \text{on } V_j.$$

Then, using $\rho_j(z) = z^h$ and $d\rho_j(z) = h z^{h-1} dz$,

$$\begin{aligned} \rho_j^*(\omega_j) &= \frac{g_j(\rho_j(z))}{(h \rho_j(z))^n} (d\rho_j(z))^n = \frac{g_j(z^h)}{(h z^h)^n} (h z^{h-1})^n (dz)^n \\ &= \frac{z^n (f[\alpha]_{2n})(z)}{h^n z^{hn}} h^n z^{hn-n} (dz)^n \\ &= (f[\alpha]_{2n})(z) (dz)^n = \lambda_j. \end{aligned}$$

And λ_j pulls back to $f(\tau)(d\tau)^n|_{U'_j}$ under δ_j .

Case 2: U_j contains a cusp s_j

Here $\delta_j(s_j) = \infty$ (so $\alpha = \delta_j^{-1} \in \text{SL}_2(\mathbb{Z})$ with $\alpha(\infty) = s_j$), and since f is meromorphic at the cusps,

$$(f[\alpha]_{2n})(z) = g_j(q_h), \quad h = \text{width of } s_j, \quad q_h = e^{2\pi i z/h},$$

where g_j is meromorphic in

$$\begin{cases} q_h \text{ at } 0 & \text{if } s_j \text{ is regular,} \\ q_h^{1/2} = e^{\pi iz/h} \text{ at } 0 & \text{if } s_j \text{ is irregular.} \end{cases}$$

Remark. The irregular case only matters in odd weight. Our weight $2n$ is even, so $f[\alpha]_{2n}[-T^h]_{2n} = f[\alpha]_{2n}[T^h]_{2n}$, where $T^h = \begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}$. Thus $f[\alpha]_{2n}$ is h -periodic either way and g_j is meromorphic in q_h itself.

The local coordinate is $q = \rho_j(z) = e^{2\pi iz/h}$. Define

$$\omega_j = \frac{g_j(q)}{(2\pi i q/h)^n} (dq)^n \quad \text{on } V_j,$$

which is meromorphic at $q = 0$. Using $\rho'_j(z) = \frac{2\pi i}{h} e^{2\pi iz/h}$,

$$\begin{aligned} \rho_j^*(\omega_j) &= \frac{g_j(e^{2\pi iz/h})}{(2\pi i e^{2\pi iz/h}/h)^n} \left(\frac{2\pi i}{h} e^{2\pi iz/h} \right)^n (dz)^n \\ &= g_j(e^{2\pi iz/h}) (dz)^n \\ &= (f[\alpha]_{2n})(z) (dz)^n = \lambda_j, \end{aligned}$$

and we get the same result as in Case 1.

Step 3 (putting this together). In both cases,

$$\psi_j^*(\omega_j) = \delta_j^*(\rho_j^*(\omega_j)) = \delta_j^*(\lambda_j) = f(\tau)(d\tau)^n \Big|_{U'_j} \quad \text{for all } j \in J,$$

so by the proposition, $\omega(f) = (\omega_j)_{j \in J}$ is a meromorphic differential on $X(\Gamma)$ that pulls back to $f(\tau)(d\tau)^n$. Writing $k = 2n$, we arrive at the main theorem of the section.

Theorem. Let $k \in \mathbb{N}$ be even and let Γ be a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$. The map

$$\omega : \mathcal{A}_k(\Gamma) \longrightarrow \Omega^{\otimes k/2}(X(\Gamma)), \quad f \longmapsto (\omega_j)_{j \in J},$$

where (ω_j) is the collection constructed above that pulls back to $f(\tau)(d\tau)^{k/2}$, is an isomorphism of complex vector spaces.

Proof sketch. Every step of the construction is linear in f , so ω is linear. It is injective since $\pi^*(\omega(f)) = f(\tau)(d\tau)^{k/2}$ recovers f . For surjectivity, let $\omega' \in \Omega^{\otimes k/2}(X(\Gamma))$. By the earlier remark, $\pi^*(\omega') = f(\tau)(d\tau)^{k/2}$ for some $f \in \mathcal{A}_k(\Gamma)$, so ω' and $\omega(f)$ have the same pullback. Each $\psi_j : U'_j \rightarrow V'_j$ is a holomorphic surjection, so ψ_j^* is injective. Hence ω'_j and

$\omega(f)_j$ agree on V'_j , which is V_j with at most one point removed, so they agree on V_j . Thus $\omega' = \omega(f)$. $\square$

An immediate corollary of this, which will be used in the next sections, is the following.

Corollary. *Let $k \in \mathbb{N}$ be even and let f be any nonzero element of $\mathcal{A}_k(\Gamma)$. Then*

$$\mathcal{A}_k(\Gamma) = \mathbb{C}(X(\Gamma)) f,$$

where $\mathbb{C}(X(\Gamma))$ denotes the field of meromorphic functions on $X(\Gamma)$. Moreover, by the isomorphism above, $\Omega^{\otimes k/2}(X(\Gamma)) = \mathbb{C}(X(\Gamma)) \omega(f)$.

Proof. If $f' \in \mathcal{A}_k(\Gamma)$, then f'/f is weakly modular of weight 0 and meromorphic at the cusps, i.e. $f'/f \in \mathcal{A}_0(\Gamma) = \mathbb{C}(X(\Gamma))$. So $f' = (f'/f) f$. Applying ω , which satisfies $\omega(gf) = g \omega(f)$ for $g \in \mathbb{C}(X(\Gamma))$, gives the second statement. $\square$

Why Should We Care? Orders of Vanishing

We now know $\mathcal{A}_k(\Gamma) \cong \Omega^{\otimes k/2}(X(\Gamma))$, so we can study the subspaces we actually care about by looking at their images

$$\omega(\mathcal{M}_k(\Gamma)) \quad \text{and} \quad \omega(\mathcal{S}_k(\Gamma)).$$

To identify these images, we compare the orders of vanishing of f and of $\omega(f)$. Let $f \in \mathcal{A}_k(\Gamma)$ and $\omega(f) = (\omega_j)_{j \in J}$. For a differential, $\nu_x(\omega)$ means the order of vanishing of ω_j in the local coordinate q at $\varphi_j(x)$. For f , we use the conventions

$$\nu_{\pi(\tau)}(f) = \frac{\nu_{\tau}(f)}{h_{\tau}} \quad \text{and} \quad \nu_{\pi(s)}(f) = \text{order of } f[\alpha]_k \text{ in } q_h \text{ at } 0,$$

where h_{τ} is the period of τ and, for the cusp $s = \alpha(\infty)$, h is its width.

For $\tau_j \in \mathcal{H}$, we have $z^{k/2}(f[\alpha]_k)(z) = g_j(z^h)$. Since $\nu_0(f[\alpha]_k) = \nu_{\tau_j}(f)$ (because α is a local biholomorphism with $\alpha(0) = \tau_j$), comparing orders gives $h \nu_0(g_j) = \frac{k}{2} + \nu_{\tau_j}(f)$. Hence

$$\nu_{\pi(\tau_j)}(\omega) \stackrel{\text{def}}{=} \nu_0\left(\frac{g_j(q)}{(hq)^{k/2}}\right) = \frac{k/2 + \nu_{\tau_j}(f)}{h} - \frac{k}{2} = \nu_{\pi(\tau_j)}(f) - \frac{k}{2} \left(1 - \frac{1}{h}\right).$$

At a nonelliptic point $h = 1$, so $\nu_{\pi(\tau_j)}(\omega) = \nu_{\pi(\tau_j)}(f)$.

For a cusp s_j , we have

$$\nu_{\pi(s_j)}(\omega) \stackrel{\text{def}}{=} \nu_0\left(\frac{g_j(q)}{(2\pi i q/h)^{k/2}}\right) = \nu_{\pi(s_j)}(f) - \frac{k}{2}.$$

Here is a small table summarizing these formulas.

2navy!5!whitewhite

Point $x \in X(\Gamma)$	Local coordinate	$\nu_x(\omega(f))$
nonelliptic $\pi(\tau)$, $\tau \in \mathcal{H}$	$q = z$	$\nu_x(f)$
elliptic $\pi(\tau)$, period h	$q = z^h$	$\nu_x(f) - \frac{k}{2} \left(1 - \frac{1}{h}\right)$
cusp $\pi(s)$, width h	$q = e^{2\pi iz/h}$	$\nu_x(f) - \frac{k}{2}$

Remark. When $k \in \mathbb{N}$ is even, the conditions above tell us exactly which differentials come from modular forms and cusp forms. A form f lies in $\mathcal{M}_k(\Gamma)$ if and only if $\nu_x(f) \geq 0$ for all $x \in X(\Gamma)$, which by the table means

$$\nu_x(\omega(f)) \geq \begin{cases} 0 & x \text{ nonelliptic,} \\ -\frac{k}{2} \left(1 - \frac{1}{h_x}\right) & x \text{ elliptic,} \\ -\frac{k}{2} & x \text{ a cusp.} \end{cases}$$

For $\mathcal{S}_k(\Gamma)$ we additionally require $\nu_x(f) \geq 1$ at the cusps, i.e. $\nu_x(\omega(f)) \geq 1 - \frac{k}{2}$ there. So we have the picture

$$\begin{array}{ccc} \mathcal{A}_k(\Gamma) & \xleftarrow{\sim} & \Omega^{\otimes k/2}(X(\Gamma)) \\ \cup & & \cup \\ \mathcal{M}_k(\Gamma) & \xrightarrow{\sim} & \omega(\mathcal{M}_k(\Gamma)) \\ \cup & & \cup \\ \mathcal{S}_k(\Gamma) & \xrightarrow{\sim} & \omega(\mathcal{S}_k(\Gamma)) \end{array}$$

and the images are cut out by *inequalities on orders of vanishing*. Divisors are the language for such inequalities, which is the subject of the next section.

Divisors and Riemann-Roch (Section 3.4)

The previous section may seem of little use for studying modular forms by itself. However, the isomorphism between automorphic forms and meromorphic differentials lets us view $\mathcal{M}_k(\Gamma)$ and $\mathcal{S}_k(\Gamma)$ as subspaces of meromorphic differentials cut out by conditions on their zeros and poles. Now we use a theorem from complex geometry (Riemann-Roch) to compute the dimensions of such subspaces. Throughout this section, X is a **compact**

Riemann surface of genus g . In the context of modular forms, we imagine $X = X(\Gamma)$ for some congruence subgroup Γ of $\mathrm{SL}_2(\mathbb{Z})$.

Divisors

Definition. A **divisor** on X is a formal sum over the points of X ,

$$D = \sum_{x \in X} n_x \cdot x, \quad n_x \in \mathbb{Z},$$

where all but finitely many n_x are 0.

Example. For $p, q \in X$ with $p \neq q$, some simple divisors are

$$D = p, \quad D = p + q, \quad D = 2p - q.$$

Notice that the set $\mathrm{Div}(X)$ of divisors on X forms the free abelian group on the points of X . That is, for divisors $D = \sum n_x \cdot x$ and $D' = \sum n'_x \cdot x$ we have

$$D + D' = \sum_{x \in X} (n_x + n'_x) \cdot x.$$

By convention, we say $D \geq D'$ if $n_x \geq n'_x$ for all $x \in X$. In particular, $D \geq 0$ means every coefficient of D is nonnegative.

Definition. The **degree** of a divisor $D = \sum n_x \cdot x$ is

$$\deg(D) = \sum_{x \in X} n_x.$$

Using the examples above, $D = p$ has degree 1, $D = p + q$ has degree 2, and $D = 2p - q$ has degree 1. It is immediate that the map $D \mapsto \deg(D)$ is a group homomorphism $\mathrm{Div}(X) \rightarrow \mathbb{Z}$. Its kernel is the subgroup of degree-zero divisors,

$$\mathrm{Div}^0(X) = \{D \in \mathrm{Div}(X) : \deg(D) = 0\}.$$

Principal Divisors

We associate to each nonzero meromorphic function on X a divisor recording its order of vanishing at each point. For $f \in \mathbb{C}(X)^*$ and $x \in X$, let $\nu_x(f)$ be the order of vanishing of f at x , computed in any local coordinate φ_j near x as $\nu_{\varphi_j(x)}(f \circ \varphi_j^{-1})$. This does not depend on the chart since transition maps are biholomorphic. Since X is compact and zeros and poles are isolated, f has only finitely many zeros and poles.

Definition. For $f \in \mathbb{C}(X)^*$, the **principal divisor** of f is

$$\operatorname{div}(f) = \sum_{x \in X} \nu_x(f) \cdot x.$$

Since $\nu_x(fg) = \nu_x(f) + \nu_x(g)$, we have $\operatorname{div}(fg) = \operatorname{div}(f) + \operatorname{div}(g)$, so $\operatorname{div} : \mathbb{C}(X)^* \rightarrow \operatorname{Div}(X)$ is another group homomorphism. Its kernel is $\mathbb{C}^*$: a function with no zeros or poles is holomorphic on the compact surface X , hence constant. The image is the subgroup of principal divisors, denoted $\operatorname{Div}_\ell(X)$.

The following proposition comes from complex analysis and is the key fact about principal divisors.

Proposition. *For every $f \in \mathbb{C}(X)^*$, $\deg(\operatorname{div}(f)) = 0$. That is, a nonzero meromorphic function on a compact Riemann surface has as many zeros as poles, counted with multiplicity.*

Proof. If f is constant, then $\operatorname{div}(f) = 0$. Otherwise, f defines a nonconstant holomorphic map $f : X \rightarrow \widehat{\mathbb{C}}$ between compact Riemann surfaces. Recall from Section 3.1 that such a map has a degree $d \geq 1$: every point of $\widehat{\mathbb{C}}$ has exactly d preimages counted with ramification degree. The preimages of 0 are the zeros of f , with ramification degree $e_x = \nu_x(f)$, and the preimages of ∞ are the poles of f , with ramification degree $e_x = -\nu_x(f)$. Hence

$$\sum_{\nu_x(f) > 0} \nu_x(f) = d = \sum_{\nu_x(f) < 0} (-\nu_x(f)),$$

and so $\deg(\operatorname{div}(f)) = d - d = 0$. □

Remark. Another way to see this is the residue theorem: the meromorphic differential df/f has residue $\nu_x(f)$ at each x , and the residues of a meromorphic differential on a compact Riemann surface sum to 0.

Thus $\operatorname{Div}_\ell(X) \subset \operatorname{Div}^0(X)$. Two divisors D, D' are called **linearly equivalent** if $D - D' \in \operatorname{Div}_\ell(X)$, i.e. $D' = D + \operatorname{div}(h)$ for some $h \in \mathbb{C}(X)^*$. Linearly equivalent divisors have the same degree. The quotient

$$\operatorname{Pic}^0(X) = \operatorname{Div}^0(X) / \operatorname{Div}_\ell(X)$$

is the **degree-zero Picard group** of X . (For a complex torus, Pic^0 turns out to be isomorphic to the torus itself.)

Divisors of Meromorphic Differentials

We can also attach divisors to meromorphic differentials. Let $\omega = (\omega_j)_{j \in J} \in \Omega^{\otimes n}(X)$ be nonzero, with $\omega_j = f_j(q_j)(dq_j)^n$. For $x \in U_j$ define

$$\nu_x(\omega) = \nu_{\varphi_j(x)}(f_j).$$

This is well-defined. By compatibility, $f_j(q_j) = f_k(\varphi_{k,j}(q_j)) \varphi'_{k,j}(q_j)^n$, and $\varphi'_{k,j}$ has no zeros since $\varphi_{k,j}$ is biholomorphic, so f_j and f_k have the same order at corresponding points.

Definition. The **divisor** of a nonzero meromorphic differential $\omega \in \Omega^{\otimes n}(X)$ is

$$\operatorname{div}(\omega) = \sum_{x \in X} \nu_x(\omega) \cdot x.$$

A **canonical divisor** on X is the divisor $\operatorname{div}(\lambda)$ of a nonzero meromorphic differential $\lambda \in \Omega^{\otimes 1}(X)$.

Every compact Riemann surface has a nonzero meromorphic differential (this is a theorem from the theory of Riemann surfaces; for $X(\Gamma)$ we built them explicitly from automorphic forms). Moreover, if $\lambda \in \Omega^{\otimes 1}(X)$ is nonzero, then

$$\Omega^{\otimes n}(X) = \mathbb{C}(X) \lambda^n.$$

Indeed, the ratio of two nonzero degree- n differentials is a meromorphic function, since the transition factors $\varphi'^n_{k,j}$ cancel. Since $\operatorname{div}(f\lambda^n) = \operatorname{div}(f) + n\operatorname{div}(\lambda)$, any two canonical divisors are linearly equivalent, and in particular they all have the same degree. This matches the Corollary of Section 3.3 when $X = X(\Gamma)$.

The Linear Space of a Divisor

Definition. For $D \in \operatorname{Div}(X)$, the **linear space** of D is

$$\mathcal{L}(D) = \{f \in \mathbb{C}(X) : f = 0 \text{ or } \operatorname{div}(f) + D \geq 0\},$$

and we write $\ell(D) = \dim_{\mathbb{C}} \mathcal{L}(D)$.

To understand the condition, write $D = \sum n_x \cdot x$. Then $\operatorname{div}(f) + D \geq 0$ means $\nu_x(f) \geq -n_x$ for all x : where $n_x > 0$, f may have a pole of order at most n_x , and where $n_x < 0$, f must vanish to order at least $-n_x$. Since $\nu_x(f+g) \geq \min(\nu_x(f), \nu_x(g))$, $\mathcal{L}(D)$ is a $\mathbb{C}$ -vector space, and it is a fact that it is finite-dimensional when X is compact.

Proposition. Let $D, D' \in \text{Div}(X)$.

- (a) $\mathcal{L}(0) = \mathbb{C}$, so $\ell(0) = 1$.
- (b) If $\deg(D) < 0$, then $\mathcal{L}(D) = 0$, so $\ell(D) = 0$.
- (c) If $D \leq D'$, then $\mathcal{L}(D) \subset \mathcal{L}(D')$.
- (d) If $D' = D + \text{div}(h)$ with $h \in \mathbb{C}(X)^*$, then $\mathcal{L}(D') = h^{-1}\mathcal{L}(D)$, so $\ell(D') = \ell(D)$.

Proof. (a) $f \in \mathcal{L}(0)$ nonzero means $\text{div}(f) \geq 0$, so f is holomorphic on compact X and hence constant. (b) If $f \neq 0$ were in $\mathcal{L}(D)$, then $\text{div}(f) + D \geq 0$ would have nonnegative degree, but $\deg(\text{div}(f) + D) = 0 + \deg(D) < 0$. (c) is immediate from the definition. (d) We have $\text{div}(g) + D' = \text{div}(gh) + D$, so $g \in \mathcal{L}(D')$ if and only if $gh \in \mathcal{L}(D)$. $\square$

By (d), $\ell(\text{div}(\lambda))$ does not depend on which canonical divisor we choose. The canonical divisor also controls the holomorphic differentials: for $\lambda \in \Omega^{\otimes 1}(X)$ nonzero and $f \in \mathbb{C}(X)$,

$$f \in \mathcal{L}(\text{div}(\lambda)) \iff \text{div}(f) + \text{div}(\lambda) \geq 0 \iff \text{div}(f\lambda) \geq 0 \iff f\lambda \text{ is holomorphic.}$$

So $f \mapsto f\lambda$ is an isomorphism from $\mathcal{L}(\text{div}(\lambda))$ to the space of holomorphic differentials of degree 1 on X .

The Riemann-Roch Theorem

We now have all the language needed for the main theorem.

Theorem (Riemann-Roch). Let X be a compact Riemann surface of genus g , and let $\text{div}(\lambda)$ be a canonical divisor on X . Then for every $D \in \text{Div}(X)$,

$$\ell(D) = \deg(D) - g + 1 + \ell(\text{div}(\lambda) - D).$$

The proof is beyond the scope of the text (see a book on Riemann surfaces, e.g. Miranda, *Algebraic Curves and Riemann Surfaces*). What makes the theorem remarkable is that it links a *topological* invariant, the genus g , to *analytic* data, the dimensions of spaces of meromorphic functions with prescribed zeros and poles. The key consequences come from plugging in well-chosen divisors.

Corollary. Let X be a compact Riemann surface of genus g and $\lambda \in \Omega^{\otimes 1}(X)$ nonzero.

- (a) $\ell(\text{div}(\lambda)) = g$. That is, the space of holomorphic differentials on X has dimension g .
- (b) $\deg(\text{div}(\lambda)) = 2g - 2$.
- (c) If $\deg(D) > 2g - 2$, then $\ell(D) = \deg(D) - g + 1$.
- (d) For $n \in \mathbb{N}$ and any nonzero $\omega \in \Omega^{\otimes n}(X)$, $\deg(\text{div}(\omega)) = n(2g - 2)$.

Proof. (a) Take $D = 0$. Since $\ell(0) = 1$, Riemann-Roch reads $1 = 0 - g + 1 + \ell(\text{div}(\lambda))$, so

$$\ell(\operatorname{div}(\lambda)) = g.$$

(b) Take $D = \operatorname{div}(\lambda)$. Using (a) and $\ell(0) = 1$, $g = \deg(\operatorname{div}(\lambda)) - g + 1 + 1$, so $\deg(\operatorname{div}(\lambda)) = 2g - 2$.

(c) By (b), $\deg(\operatorname{div}(\lambda) - D) = 2g - 2 - \deg(D) < 0$, so $\ell(\operatorname{div}(\lambda) - D) = 0$ and Riemann-Roch reduces to $\ell(D) = \deg(D) - g + 1$.

(d) Write $\omega = f\lambda^n$ with $f \in \mathbb{C}(X)^*$. Then $\deg(\operatorname{div}(\omega)) = \deg(\operatorname{div}(f)) + n \deg(\operatorname{div}(\lambda)) = 0 + n(2g - 2)$. $\square$

Examples

Example (The Riemann sphere, $g = 0$). Cover $\widehat{\mathbb{C}}$ by the charts z on $\mathbb{C}$ and $w = 1/z$ near ∞ . Consider $\lambda = dz$. On $\mathbb{C}$ it has no zeros or poles. Near ∞ , the transition map is $z = 1/w$, so

$$dz = d(1/w) = -w^{-2} dw,$$

and $\nu_\infty(dz) = -2$. Hence $\operatorname{div}(dz) = -2 \cdot \infty$ has degree $-2 = 2(0) - 2$, consistent with $g = 0$.

Now take $D = m \cdot \infty$ with $m \geq 0$. The meromorphic functions on $\widehat{\mathbb{C}}$ are the rational functions, and $f \in \mathcal{L}(D)$ means f has no poles in $\mathbb{C}$ and a pole of order at most m at ∞ . So $\mathcal{L}(D)$ is the space of polynomials of degree at most m , and $\ell(D) = m + 1 = \deg(D) - g + 1$, as part (c) predicts (since $m > -2$).

Example (The complex torus, $g = 1$). Return to $X = \mathbb{C}/\Lambda$ and the differential dz from Section 3.3. We saw it is holomorphic with no zeros, so $\operatorname{div}(dz) = 0$ and $\deg(\operatorname{div}(dz)) = 0 = 2(1) - 2$, consistent with $g = 1$. By part (a), the holomorphic differentials form the 1-dimensional space $\mathbb{C} dz$.

For a single point $D = p$, we have $\deg(D) = 1 > 0 = 2g - 2$, so $\ell(p) = 1 - 1 + 1 = 1$ and $\mathcal{L}(p) = \mathbb{C}$. In other words, *there is no meromorphic function on a torus with exactly one simple pole*. For $D = 2 \cdot 0$ we get $\ell(D) = 2$, and indeed $\mathcal{L}(2 \cdot 0)$ is spanned by 1 and the Weierstrass $\wp$ -function $\wp_\Lambda$, which has a double pole at the lattice points.

Looking Ahead

Remark. Combining the two sections, for $f \in \mathcal{A}_k(\Gamma)$ nonzero with k even, part (d) of the corollary applied to $\omega(f) \in \Omega^{\otimes k/2}(X(\Gamma))$ gives

$$\deg(\operatorname{div}(\omega(f))) = \frac{k}{2}(2g - 2) = k(g - 1),$$

where g is the genus of $X(\Gamma)$. Together with the table of orders of vanishing from

Section 3.3, the conditions defining $\omega(\mathcal{M}_k(\Gamma))$ and $\omega(\mathcal{S}_k(\Gamma))$ become conditions of the form $\operatorname{div}(\omega) \geq -D$ for an explicit divisor D . Riemann-Roch then computes the dimensions. In Section 3.5 this yields, for $k \geq 2$ even,

$$\dim \mathcal{M}_k(\Gamma) = (k-1)(g-1) + \left\lfloor \frac{k}{4} \right\rfloor \varepsilon_2 + \left\lfloor \frac{k}{3} \right\rfloor \varepsilon_3 + \frac{k}{2} \varepsilon_\infty,$$

where $\varepsilon_2, \varepsilon_3, \varepsilon_\infty$ are the numbers of elliptic points of period 2, of period 3, and of cusps of $X(\Gamma)$.